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Department of Computer Science and Engineering (Data Science)

MAJOR PROJECT PHASE -I ABSTRACT SUBMISSION FORM (2026–27)

Semester: VII **Section:** A

Project Group ID : CSDMP230013

ENHANCING ADVERSARIAL ROBUSTNESS: A NOVEL BAYESIAN UNCERTAINITY-BASED ENSEMBLE LEARNING METHOD

Abstract:

Deep Learning (DL) models are highly vulnerable to adversarial attacks, where small input perturbations can significantly affect predictions and reduce system reliability. Existing ensemble learning approaches improve robustness but often fail to dynamically evaluate model reliability, assign appropriate weights, and protect the final aggregation model. To address these limitations, this project proposes a Bayesian Uncertainty-Based Ensemble Learning (BUEL) method. The approach estimates Bayesian uncertainty for each model prediction and dynamically assigns weights based on model reliability. More reliable models contribute more significantly to the final prediction, while weights can be updated using observed data. Adversarial training is also applied at the aggregation level to strengthen the overall ensemble against sophisticated attacks. The system includes data preprocessing, model training, Bayesian uncertainty estimation, ensemble learning, adversarial training, and evaluation modules. Experiments on offensive language detection demonstrate improved adversarial robustness while maintaining reliable predictive performance. The project supports **SDG 9** (Industry, Innovation and Infrastructure) by promoting resilient AI technologies and **SDG 16** (Peace, Justice and Strong Institutions) through improved security and reliability of intelligent systems. Overall, BUEL provides an adaptive and scalable approach for developing robust deep learning applications.

Keywords: Adversarial Robustness, Bayesian Uncertainty, Ensemble Learning, Deep Learning, Adversarial Attacks, Dynamic Weight Assignment, Model Reliability, Secure Artificial Intelligence.

Supervisor Approval

I hereby approve the above abstract for Major Project development.

Name of the Supervisor: Mr. Md Ashique Hussain

Signature: _____ **Date**: _____

Coordinator **Head of the Department**